

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I need a btb connector material

Assistant: To help identify the best material for your BTB connector, could you please clarify:

1. What is the exact use of the polymer in your BTB connector? For example, is it for the housing, insulator body, latching mechanism, or another component?
2. Where will this connector be used in your solution? For instance, is it for a laptop, server, military device, or another end-use application?

Your answers will help me provide more targeted material suggestions.

User: housing,laptops and cell phones.

Assistant: Thank you for clarifying. To further narrow down the material options, could you please provide:

1. What are the typical operating conditions for your connector housing? For example, is it exposed to elevated temperatures from nearby electronics, or is it mostly at room temperature inside the device?
2. What is the minimum wall thickness of your connector housing?
3. Do you have any preferred base materials for this application (such as specific polymers you've used before)?
4. Are you open to alternative material suggestions if they offer improved performance

or cost benefits?

User: i dont have that information and i dont have any preference

Assistant: Here are the key material requirements (CTQs) assumed for your BTB connector housing in laptops and cell phones:

1. UL 94 V-0 flammability rating at 0.4 mm minimum thickness
2. Lead-free soldering resistance (withstands reflow peaks up to 260-275°C)
3. Injection molding suitability
4. High flow for thin-wall, fine-pitch features
5. Dimensional stability and low warpage
6. Relative Thermal Index (Electrical) $\geq 130^{\circ}\text{C}$
7. Low relative permittivity and low dissipation factor (for signal integrity)

Are all of these requirements correct for your application? Please confirm if these match your needs, or let me know if any should be changed.

User: looks good recommend the grades based on these ctq

Assistant: Based on your requirements, here are the top grade suggestions for BTB connector housing in laptops and cell phones:

1. ZENITE® 251 – Excellent dielectric properties ($D_k \sim 2.7$), UL 94 V-0 at 0.15 mm, high flow, dimensional stability, and proven reflow resistance.
2. VECTRA® S471 – UL 94 V-0 at 0.38 mm, high flow, low warpage, RTI Elec 130°C, designed for fine-pitch connector housings.
3. VECTRA® T130 – UL 94 V-0 at 0.40 mm, high flow, RTI Elec $\geq 130^{\circ}\text{C}$, suited for thin-wall E/E parts.
4. VECTRA® E130i – UL 94 V-0 at 0.20 mm, excellent flow, widely used for fine-pitch

connectors, RTI Elec 130°C.

5. VECTRA® S475 – UL 94 V-0 at 0.10 mm, super high flow, low warpage, RTI Elec 130°C.

6. VECTRA® E471I – UL 94 V-0 at 0.15 mm, high flow, proven for fine-pitch connectors.

7. VECTRA® E130i ECO-B 352 – Sustainable variant of E130i, same performance, bio content.

8. VECTRA® S475 ECO-B 152 – Sustainable variant of S475, same performance, bio content.

These grades may be suitable options for your application, but final material selection and product design decisions should be based on your specific requirements and testing. If your project involves critical safety or regulatory compliance, please ensure all applicable standards are met. Celanese does not manufacture or certify products for these applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.